

UNets, part 1

Weekend 3, Friday afternoon

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CAS Building ML/AI Applications, ETH Zurich

Friday 2 October 2026

Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

Segmentation and
detection



Generation



Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

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Feature extraction



Cat color: black

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Cat's eye color:
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Dog's eye color:
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Segmentation and
detection



Generation

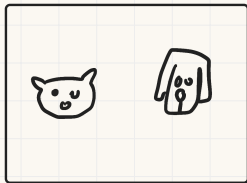


1

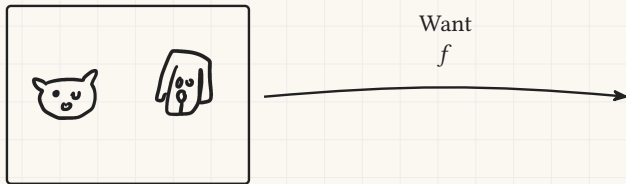
Section 1

Segmentation

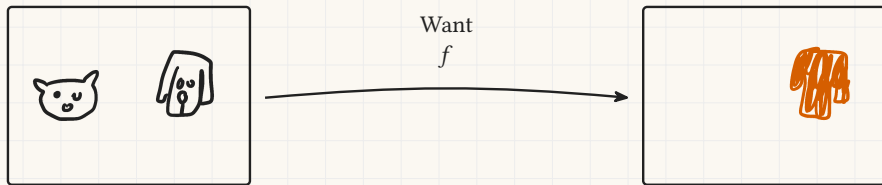
The problem of segmentation



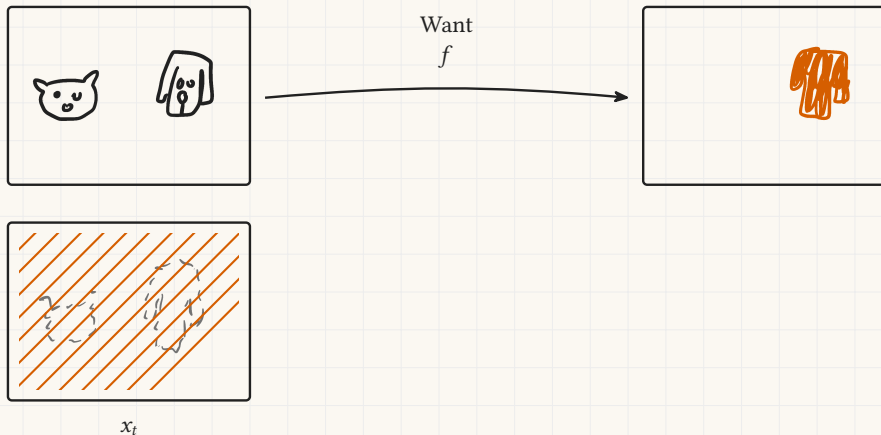
The problem of segmentation



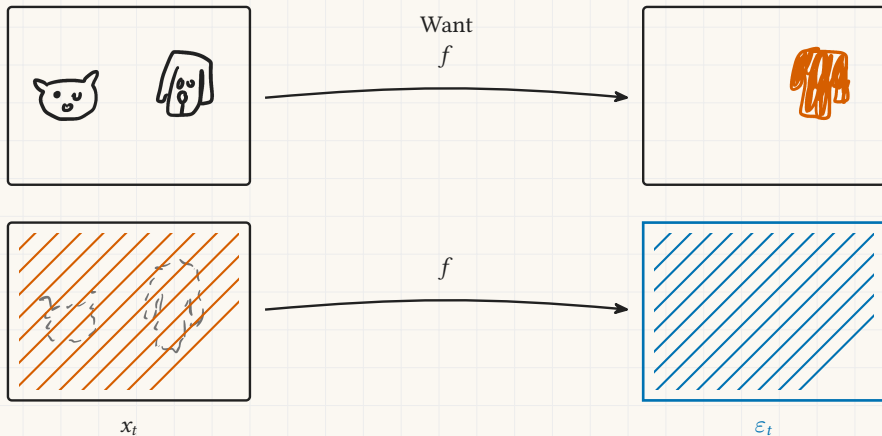
The problem of segmentation



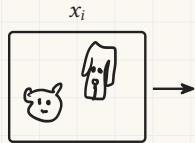
The problem of segmentation



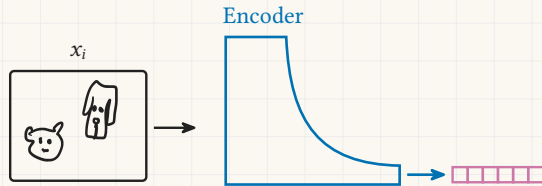
The problem of segmentation



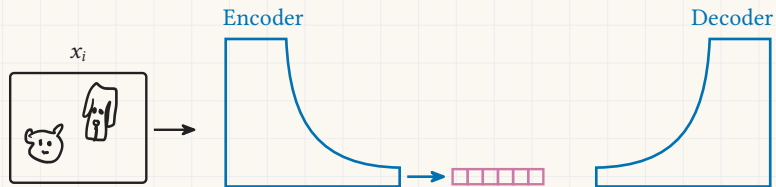
Strategy



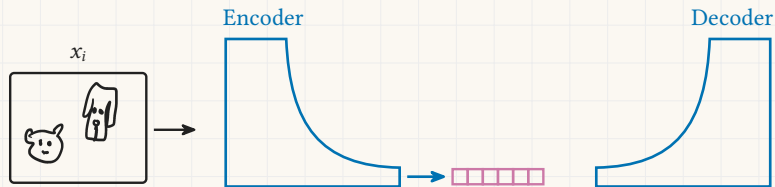
Strategy



Strategy



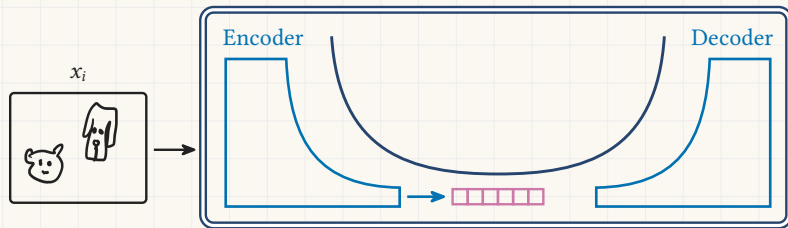
Strategy



2 tasks

- 1) Reconstruct
- 2) Identify object of interest
(e.g. dogs or noise)

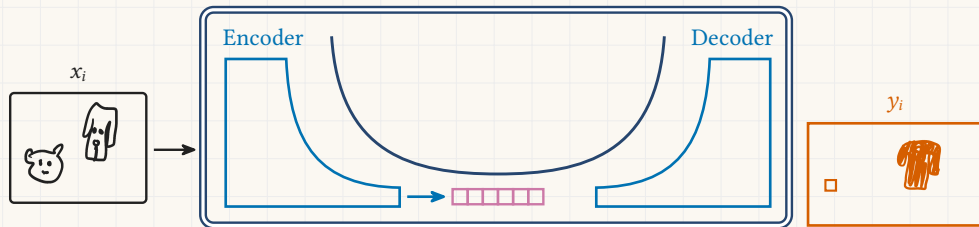
Strategy



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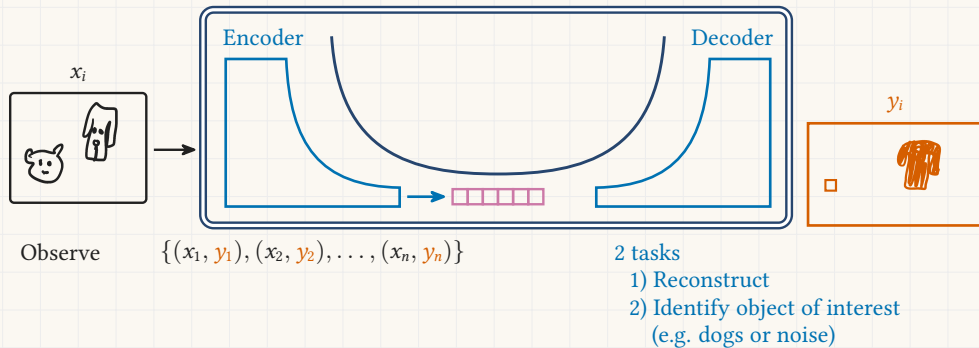
Strategy



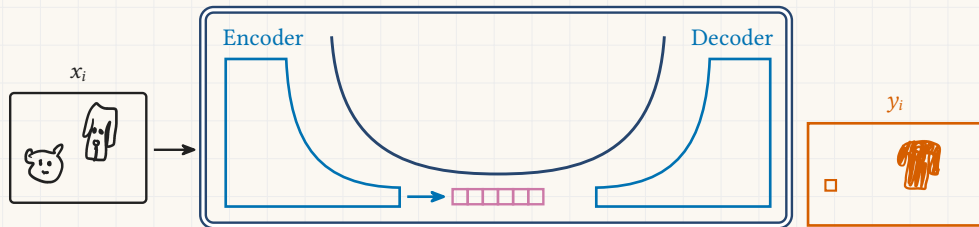
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Strategy



Strategy

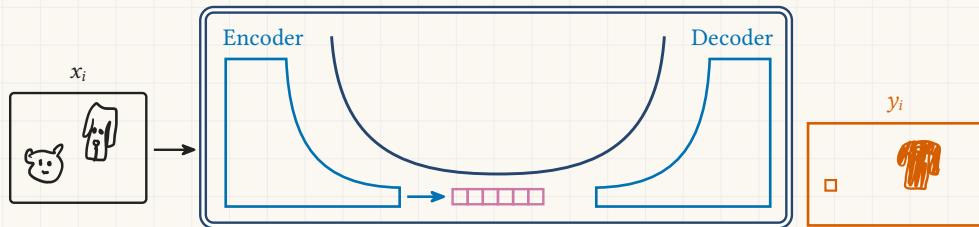


Observe $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$
Hypothesize $y = \text{UNet}(x)$

2 tasks

- 1) Reconstruct
- 2) Identify object of interest
(e.g. dogs or noise)

Strategy



Observe
Hypothesize
Validate

$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

$y = \text{UNet}(x)$

do

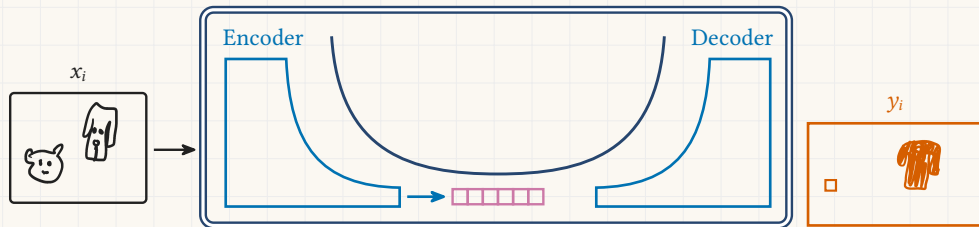
for $i \leq n$:

$\hat{y}_i = \text{UNet}(x_i)$

2 tasks

- 1) Reconstruct
- 2) Identify object of interest
(e.g. dogs or noise)

Strategy



Observe
Hypothesize
Validate

$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

$y = \text{UNet}(x)$

do

for $i \leq n$:

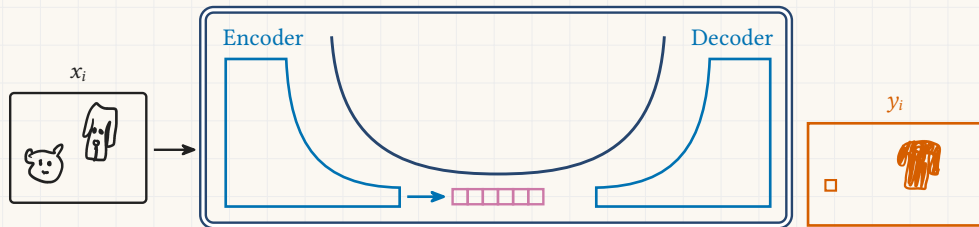
$\hat{y}_i = \text{UNet}(x_i)$

$l_i = \sum_{jk} (\hat{y}_{ijk} - y_{ijk})^2$

2 tasks

- 1) Reconstruct
- 2) Identify object of interest
(e.g. dogs or noise)

Strategy



Observe
Hypothesize
Validate

$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

$y = \text{UNet}(x)$

do

for $i \leq n$:

$\hat{y}_i = \text{UNet}(x_i)$

$l_i = \sum_{jk} (\hat{y}_{ijk} - y_{ijk})^2$

$l = \frac{1}{n} \sum l_i$

$\hat{w} \leftarrow \hat{w} - \eta \nabla l$

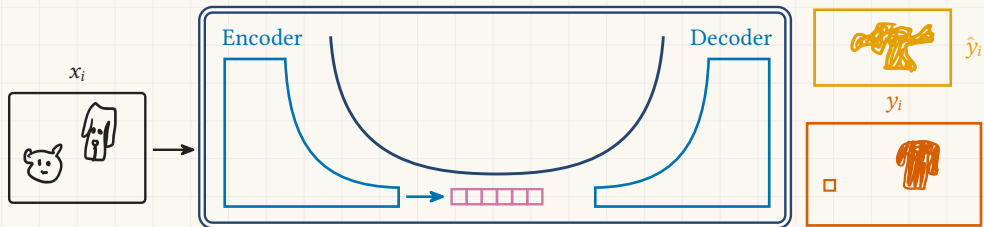
until ...

2 tasks

1) Reconstruct

2) Identify object of interest
(e.g. dogs or noise)

Strategy



Observe
Hypothesize
Validate

$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

$y = \text{UNet}(x)$

do

for $i \leq n$:

$\hat{y}_i = \text{UNet}(x_i)$

$l_i = \sum_{j,k} (\hat{y}_{ijk} - y_{ijk})^2$

$l = \frac{1}{n} \sum l_i$

$\hat{w} \leftarrow \hat{w} - \eta \nabla l$

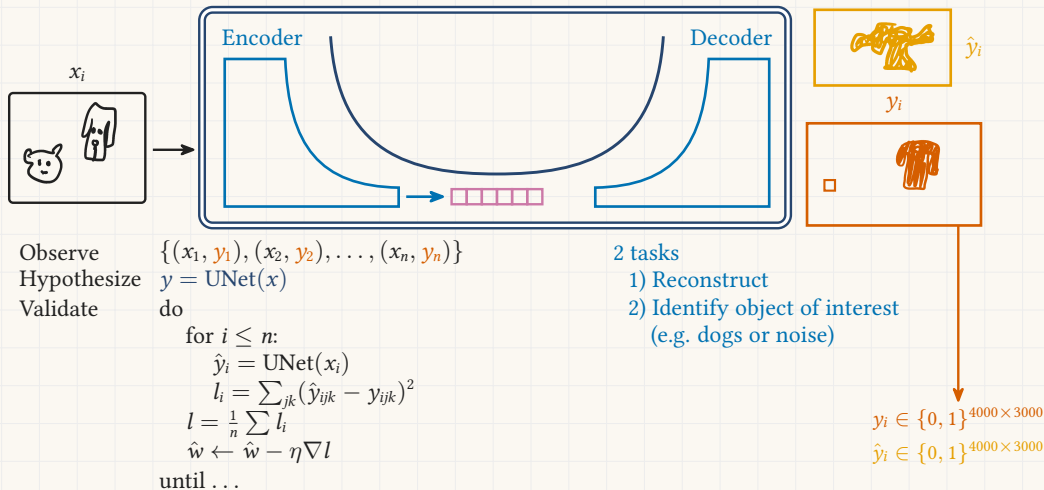
until ...

2 tasks

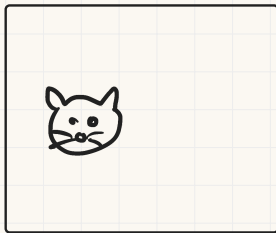
1) Reconstruct

2) Identify object of interest
(e.g. dogs or noise)

Strategy



Real segmentation



y_i

Real segmentation



y_i

Real segmentation



y_i

y_{ijk} { x_{ijk} if dog not there

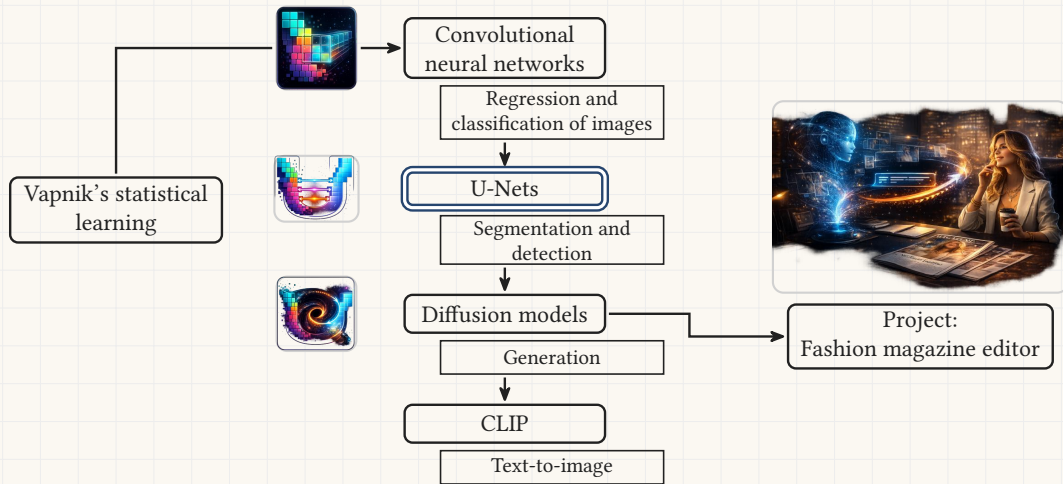
Real segmentation



y_i

y_{ijk} { x_{ijk} if dog not there
"red" if dog is there

Agenda

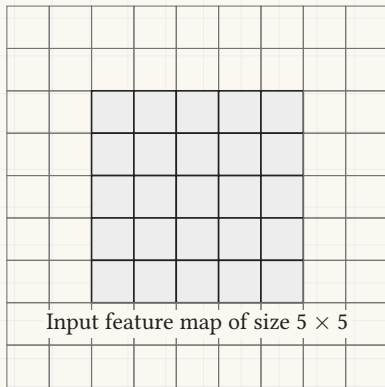


2

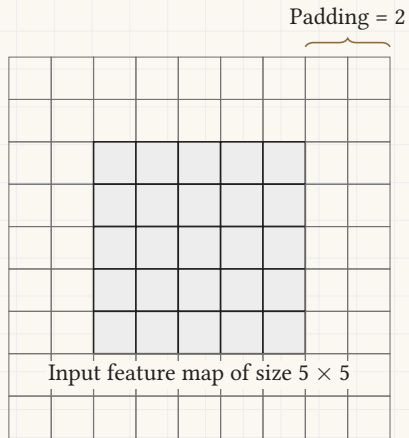
Section 2

Convolutions and deconvolutions

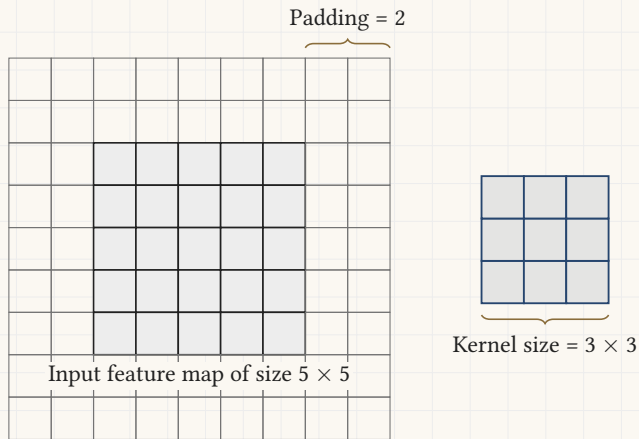
Convolutional filters



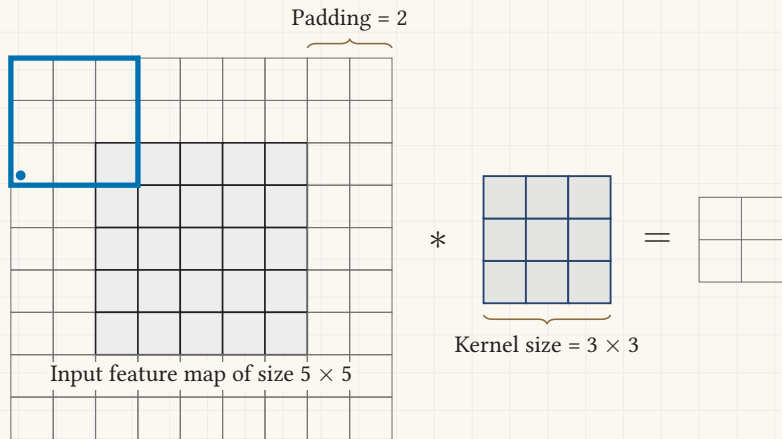
Convolutional filters



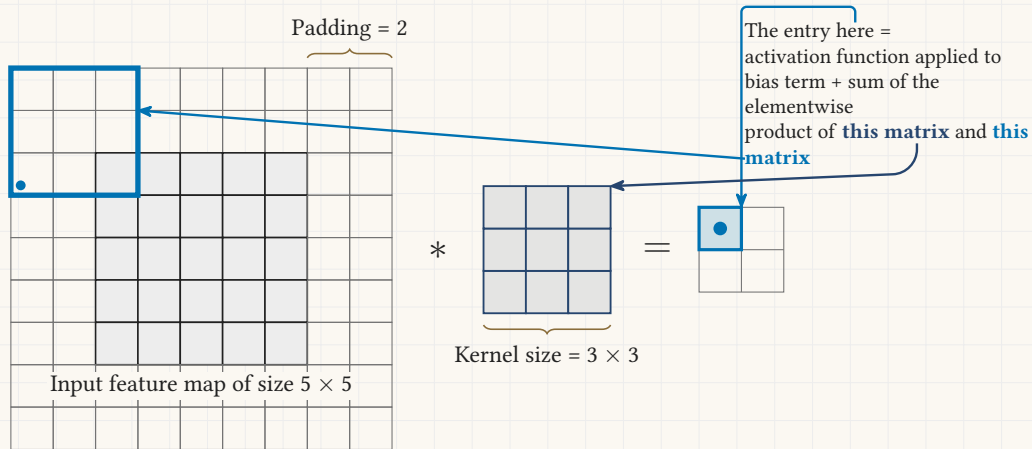
Convolutional filters



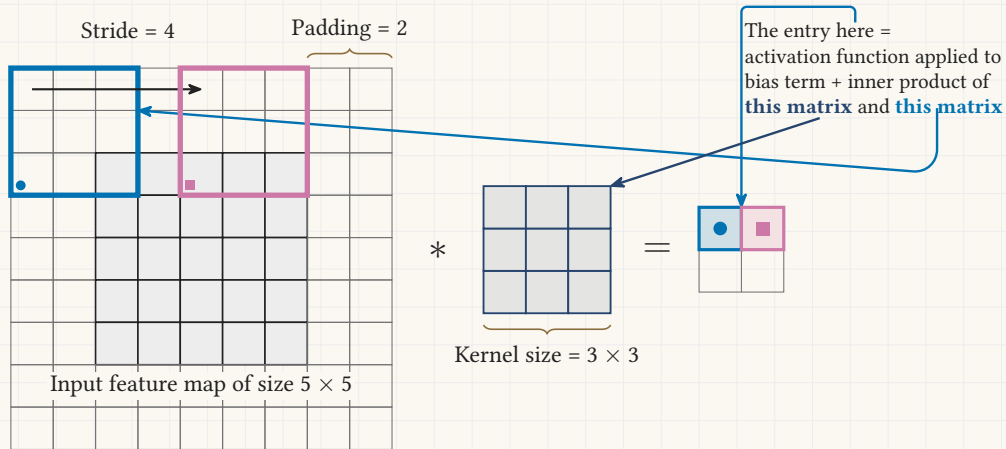
Convolutional filters



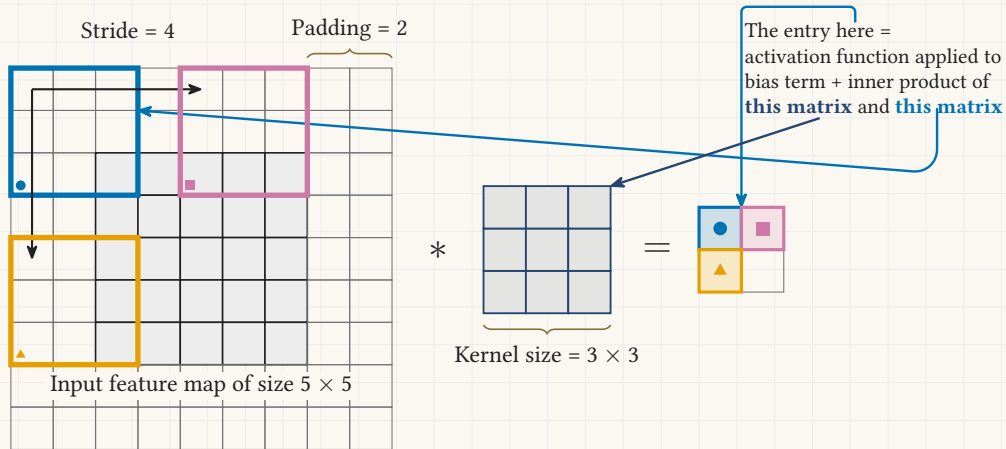
Convolutional filters



Convolutional filters



Convolutional filters



Convolutional filters

